

Chapter 15: AC Distribution, TRUs, Inverters & Architecture

Syllabus Source: CAE Oxford (pp. 139-166 (Elec)) | Pass Target: >= 90% AviationExam

1. BUSBAR ARCHITECTURES: SPLIT-BUS VS PARALLEL SYSTEMS

Commercial aircraft distribute electrical power through rigid copper or aluminum conductor rails called busbars (buses). Two distinct system philosophies govern AC generator connection:

Distribution System	Operating Philosophy & Generator Connection	Aircraft Examples & Nuances
Split-Bus System (Isolated)	Generators NEVER operate in parallel! Each AC generator powers its own independent AC busbar.	Standard on twin-engine aircraft (A320, B737, B777, B787). If one generator fails, a Bus Tie Breaker (BTB) closes automatically so the remaining generator powers both sides. Paralleling is strictly prohibited!
Parallel System	All AC generators are synchronized in frequency, voltage, and phase angle, connected to a common tie busbar.	Four-engine heavy transports (B747-400, older widebodies). Requires complex real (kW) and reactive (kVAR) load sharing circuitry in the GCU.

2. POWER CONVERSION UNITS: TRUS & STATIC INVERTERS

Modern airliners carry both AC and DC equipment, requiring two-way electrical power conversion:

- Transformer Rectifier Unit (TRU): Converts primary 115V AC (400 Hz) into 28V DC (nominal) to power DC busbars, charge batteries, and supply DC avionics. Contains a step-down transformer and a 6-phase bridge silicon diode rectifier. Has NO MOVING PARTS and requires cooling airflow.
- Static Inverter: Converts 28V DC (from aircraft batteries or DC buses) into 115V AC (single-phase, 400 Hz) to power essential flight instruments (standby attitude indicator, flight controls, navigation radios) during complete AC generator failure.

[!] EXAM TRAP (AVIATIONEXAM) - TRU vs Inverter Conversion Direction Trap

- TRU (Transformer Rectifier Unit): Converts AC -> DC (115V AC in -> 28V DC out)! - Static Inverter: Converts DC -> AC (28V DC in -> 115V AC out)! - Mnemonic: Inverter INVERTS the normal flow (takes battery DC and produces AC)!

3. GENERATOR CONTROL UNIT (GCU) & AUTOMATIC LOAD SHEDDING

Each AC generator is supervised by an electronic Generator Control Unit (GCU) providing protection and switching:

- GCU Protective Trips: Monitors for Overvoltage, Undervoltage, Overfrequency, Underfrequency, Feeder Fault (differential current between neutral and bus), and Reverse Power. If a fault occurs, the GCU opens the Generator Breaker (GB) and de-excites the generator field.

- Automatic Load Shedding: If an electrical generator is lost in flight, the remaining generator may exceed 100% rated capacity. The electrical system automatically sheds non-essential utility loads (galley ovens, cabin entertainment, passenger seat power, secondary water heaters) to preserve electrical power for flight instruments and safety systems!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): AC Split-Bus & Load Shedding S

Rule 1: Engine generators on modern jets are NEVER operated in parallel in flight. Rule 2: Bus Tie Breakers (BTBs) close automatically if a generator fails, allowing the remaining generator or APU to power both buses. Rule 3: Automatic Load Shedding: If one generator carries the entire aircraft, non-essential loads (Main Galleys, passenger entertainment, utility heaters) are shed first to protect essential flight instrumentation.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the operational function of a Transformer Rectifier Unit (TRU) in an aircraft electrical system?

- [A]: To convert 28V DC into 115V AC
- [B]: To convert 115V AC into 28V DC
- [C]: To regulate generator frequency at 400 Hz
- [D]: To charge oxygen cylinders

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A TRU steps down 115V AC power and rectifies it through solid-state silicon diodes into 28V DC power for the aircraft's DC busbars.

Q2: In a twin-engine jet aircraft featuring a 'split-bus' electrical architecture, what prevents two AC generators from operating in parallel?

- [A]: A reverse current relay
- [B]: Interlocked Bus Tie Breakers (BTBs) that ensure generators operate completely isolated and never in parallel
- [C]: Static inverters
- [D]: Thermal fuses

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In split-bus systems, parallel connection is prohibited. Control logic interlocks ensure that when an external generator or second engine generator connects, the bus tie opens or isolates.

Q3: Following an in-flight generator failure on a commercial airliner, what occurs automatically during 'load shedding'?

- [A]: Fuel is jettisoned from the wings
- [B]: Non-essential electrical consumers (galley equipment, passenger entertainment) are disconnected automatically to prevent overloading the remaining generator
- [C]: The APU shuts down
- [D]: Flight controls switch to manual reversion

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Load shedding disconnects high-draw commercial utility loads (galleys) to guarantee sufficient capacity for essential flight guidance, instruments, and pumps.